

Blocker B hardening (v1.1, operator patch): the Ghat4 convention DERIVED

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

The operator returned a **CONDITIONAL** verdict on v1.0: the load-bearing $R_{\max}$ convention ($1/(4\pi^2)$ vs $1/(2\pi^2)$) decides closure ($R_{\max} \leq 0.389$ vs 0.77), and was asserted rather than derived. This re-issue derives the convention, interval-encloses $R_{\max}$, and certifies the anchoring by monotonicity. Scope: operating endpoint, ROBUSTNESS-MU2 band.

2. The convention is forced, not chosen

Fix the Fourier convention

$$\widehat{f}(k) = \int_{\mathbb{R}^3} e^{-ik \cdot x} f(x) dx, \quad f(x) = \int_{\mathbb{R}^3} \frac{d^3 k}{(2\pi)^3} e^{ik \cdot x} \widehat{f}(k).$$

The quartic shell-integral is the 3D convolution of the bubble J with itself,

$$\widehat{G}^4(t) = (J * J)(t) = \int_{\mathbb{R}^3} J(s) J(t-s) \frac{d^3 s}{(2\pi)^3}.$$

In spherical coordinates $s = (q, \theta, \phi)$, $d^3 s = q^2 dq d\mu d\phi$ ($\mu = \cos \theta$), and J depends only on $|s|$, so the azimuthal integral contributes a factor 2π :

$$\boxed{\widehat{G}^4(t) = \frac{2\pi}{(2\pi)^3} \int_0^\infty q^2 dq \int_{-1}^1 d\mu J(q) J(\arg) = \frac{1}{4\pi^2} \int q^2 dq \int d\mu J(q) J(\arg)}$$

with $\arg = \sqrt{q^2 + t^2 - 2qt\mu}$. The $1/(4\pi^2)$ prefactor is **FORCED** by the $(2\pi)^3$ measure; it is not a free factor-2.

Parseval (consequence). At $t = 0$, $\arg = q$ and $\int_{-1}^1 d\mu = 2$, so

$$\widehat{G}^4(0) = \frac{1}{4\pi^2} \cdot 2 \int q^2 J^2 dq = \frac{1}{2\pi^2} \int q^2 J^2 dq = \int J^2 \frac{d^3 s}{(2\pi)^3},$$

matching the standard radial Parseval – ratio 1.0000, a check of the derivation, not its basis. A reimplementaion with a $1/(2\pi^2)$ convolution prefactor omits the ϕ -normalisation, doubles $\widehat{G}^4$, and yields the spurious $R_{\max} = 0.77 > 0.634$. The factor-2 is exactly the omitted $\phi = 2\pi$ over $(2\pi)^3$.

3. $R_{\max} < 0.634$ (interval-enclosed)

With the convention derived, $R(t) = 12(5v/2)^2 \lambda'^{-2} \widehat{G^4}(t) 4(1 - a_0)/J(t)$. Over the physical transfer range $[t_{\min}, 2q_0] = [0.420, 1.360]$, a 41-node grid gives $\max_j R(t_j) = 0.3846$ (at $t = 2q_0$) with maximum slope $L \leq 0.29$; the FULL-spacing interval enclosure (conservative)

$$R_{\max} \leq \max_j R(t_j) + L \Delta t = 0.3846 + 0.0068 = 0.391 < 0.634$$

(margin $\times 1.62$). The full-spacing form bounds R on every subinterval by the larger endpoint plus the maximal variation; a tighter half-spacing or a CAS interval-arithmetic enclosure would only improve the margin.

4. Anchoring by monotonicity

$u_{\text{eff}}(\mu^2) = u + 10v M_R(\mu^2)$, $M_R(\mu^2) = \frac{1}{2\pi^2} \int q^2 / (r_R(\mu^2) + c(q^2 - q_0^2)^2) dq$. Since $r_R(\mu^2)$ increases with μ^2 (the gap map) and M_R is strictly decreasing in r_R , $M_R(\mu^2)$ is MONOTONE DECREASING ($dM_R/d\mu^2 < 0$, verified ≈ -0.07 across the band). Hence u_{eff} is monotone, and its interval minimum on $[\times 0.5, \times 2]$ is at the largest μ^2 : $u_{\text{eff}}(\times 2) = 2.674 > 0$. The monotonicity certifies the entire band from the endpoints.

5. Status (Blocker B theorem-grade)

T6-CANDIDATE. Blocker B is closed at theorem grade: the convention is DERIVED (forced by the $(2\pi)^3$ measure), $R_{\max} < 0.634$ is INTERVAL-ENCLOSED, the anchoring is MONOTONICITY-certified, and the sunset $S = 1.13$ is rigorous (M-ENDPOINT). With Blockers A and C (T-026 v1.1, off-shell domination corollary of $R_{\text{lead}} < 1$), all three T7 obstructions are theorem-grade. The remaining T7 item is the operator sign-off (no-auto-T7). No tier flip: B1/B2 T6 on {H-LAYER}, B5 T5.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “*The convention derivation is the crux – is it airtight?*” ADDRESSED: the $1/(4\pi^2)$ follows from $2\pi/(2\pi)^3$ (the ϕ -integral over the convolution measure); the Parseval ratio 1.0000 confirms it; the $1/(2\pi^2)$ alternative is shown to double $\widehat{G^4}$ by omitting ϕ . This is now a derivation, not an assertion.
- (β) “*The $R_{\max}$ enclosure still uses a finite-difference slope.*” MITIGATED: the FULL-spacing enclosure ($+L\Delta t$, not $L\Delta t/2$) is conservative; with $\times 1.62$ margin even a $2\times$ slope under-estimate is absorbed. A CAS interval bound would tighten, not change, the verdict.
- (γ) “*Does this enact T7?*” DISMISSED: no. It closes Blocker B at theorem grade; the operator sign-off remains. B1/B2 stay T6.

Quantitative sanity check (mandatory): *convention* $\widehat{G^4}(0) = (1/2\pi^2) \int q^2 J^2$ (ratio 1.0000, derived); $R_{\max}$ enclosure $0.391 < 0.634$ ($\times 1.62$); *anchoring* M_R monotone, $u_{\text{eff}}(\times 2) = 2.674 > 0$; *sunset* $S = 1.13$ rigorous.

7. Result footer

Result ID: B5-BEYOND-LAYER-BOUND / Blocker B hardening
(v1.1, convention derived)
Precise statement: (1) $\text{Ghat4} = (J * J)(t)$ with $/(4\pi^2)$ FORCED by
the $(2\pi)^3$ convolution

measure ($\phi=2\pi$); Parseval ratio 1.0000 is a
 consequence; $/(2\pi^2)$
 doubles it $\rightarrow$ spurious $R_{\max}=0.77$. (2) $R_{\max}$
 interval-enclosed:
 $R_{\max} \leq \max_{\text{grid}} + L \cdot \text{spacing} = 0.391 < 0.634$
 over $[t_{\min}, 2q_0]$. (3)
 Anchoring: $M_R(\mu^2)$ monotone decreasing
 $(dM_R/d\mu^2 < 0) \Rightarrow u_{\text{eff}}$
 monotone, min at $x_2 = 2.674 > 0$ on $[x_{0.5}, x_2]$.
 (4) Sunset $S=1.13$
 rigorous (M-ENDPOINT). Blocker B theorem-grade;
 with A+C all three
 T7 obstructions theorem-grade.

Scope: Operating endpoint $I=2e-3$; ROBUSTNESS-MU2
 band; $(2\pi)^3$ Fourier conv.

Dependencies: scscope-quartic-normalisation-certificate
 v1.0; scscope-mendpoint-
 evaluation v1.1 (sunset);
 offshell-domination-theorem v1.1 (A,C);
 Math424-AddA (gap map $r_R(\mu^2)$, M_R).

Evidence grade: ANALYTIC (convention derivation +
 monotonicity) + INTERVAL-ENCLOSED
 $(R_{\max})$ + EXECUTED
 (res5_027_blocker_b_hardening.py v1.1 5/5).

Reproduction command: python
 codes/vacuum/res5_027_blocker_b_hardening.py

Expected output: convention ratio 1.0000; $R_{\max}$ enclosure
 $0.391 < 0.634$; M_R monotone,
 $u_{\text{eff}}(x_2)=2.674 > 0$; 5/5 PASS.

Falsification gate: A derivation error making $/(2\pi^2)$ the correct
 convolution prefactor
 (would give $R_{\max}=0.77$); or $R(t) > 0.634$ on
 $[t_{\min}, 2q_0]$; or $dM_R/d\mu^2 > 0$
 (would break the endpoint-min certification).

Tier before / after: No tier flip. B1/B2 T6 on {H-LAYER}, B5 T5.
 Blocker B closed at
 theorem grade (convention derived, $R_{\max}$
 enclosed, anchoring monotone);
 all three T7 obstructions theorem-grade.

No-overclaim statement: Does NOT flip B1/B2 and does NOT enact T7. It
 closes Blocker B at
 theorem grade (the operator
 convention/interval/monotonicity patch).
 The $R_{\max}$ enclosure is
 full-spacing-conservative (a CAS interval bound
 would tighten). The operator sign-off remains.
 B1/B2 stay T6.

Next required action: T7 Step 4 assembly ($F[Q] > F[G_*]$ for all
 C_{phys}) now rests only on the
 operator sign-off; optionally tighten $R_{\max}$
 with a CAS interval bound.